



Cloud Computing

Agenda

- **Cloud Computing Concept**
- **Characteristics of Cloud Computing**
- **Challenges of Cloud Computing**
- **Types of Cloud Computing Services**
- **Responsibilities in Cloud**
- **Cloud Deployment Models**
- **Five major actors in Cloud Computing**
- **Cloud Service Providers**
- **Cloud Computing Threats**
- **Cloud Security Considerations.**
- **Cloud Jobs**

Cloud Computing Concept

- Cloud computing delivers various types of services and applications over the Internet.
- Cloud computing is an on-demand delivery of IT capabilities
- IT infrastructure and applications are provided to subscribers as metered services over networks.
- Examples of cloud solutions include Gmail, Facebook, Dropbox...



Characteristics of Cloud Computing

- On-demand self-service
- Distributed storage
- Rapid elasticity(scalability)
- Automated management
- Broad network access
- Resource pooling
- Measured service
- Virtualization technology



Challenges of Cloud Computing

- Limited control
- Security, privacy
- Dependence on network connections
- Potential vulnerability to attacks as every component is online
- Difficulty in migrating from one service provider to another

Disadvantages

- May be Slow
- Internet Connectivity
- Agreement
- Security Issues

Types of Cloud Computing Services

SYS ADMINS

Infrastructure-as-a-Service (IaaS)

- Provides **virtual machines** and other abstracted hardware and operating systems which may be **controlled through a service API**
- E.g., Amazon EC2, Microsoft OneDrive, or Rackspace

DEVELOPERS

Security-as-a-Service (SECaaS)

- Provides **penetration testing, authentication, intrusion detection, anti-malware, security incident, and event management services**
- E.g., eSentire MDR, Switchfast Technologies, OneNeck IT Solutions, or Foundstone Managed Security Services

DEVELOPERS

Platform-as-a-Service (PaaS)

- Offers **development tools, configuration management, and deployment platforms** on-demand that can be used by subscribers to **develop custom applications**
- E.g., Google App Engine, Salesforce, or Microsoft Azure

END CUSTOMERS

Container-as-a-Service (CaaS)

- Offers **virtualization of container engines**, and management of containers, applications, and clusters, through a web portal or API
- E.g., Amazon EC2, or Google Kubernetes Engine (GKE)

END CUSTOMERS

Software-as-a-Service (SaaS)

- Offers **software to subscribers** on-demand **over the Internet**
- E.g., web-based office applications like Google Docs or Calendar, Salesforce CRM, or Freshbooks

END CUSTOMERS

Function-as-a-Service (FaaS)

- Provides a platform for developing, running, and managing **application functionalities for microservices**
- E.g., AWS Lambda, Google Cloud Functions, Microsoft Azure Functions, or Oracle Functions

SYS ADMINS

Identity-as-a-Service (IDaaS)

- Offers **IAM services** including SSO, MFA, IGA, and intelligence collection
- E.g., OneLogin, Centrify Identity Service, Microsoft Azure Active Directory, or Okta

END CUSTOMERS

Anything-as-a-Service (XaaS)

- Offers anything as a service over the **Internet based on the user's demand** like digital products, food, transportation, medical consultations, etc.
- E.g., Salesforce, AWS, Google Compute Engine, Azure, O365 and G Suite, JumpCloud

Types of Cloud Computing Services

- **Infrastructure-as-a-Service (IaaS)**
 - This cloud computing service enables subscribers to use on-demand fundamental IT resources, such as computing power, virtualization, data storage, network and OS.
- **Platform-as-a-Service (PaaS)**
 - This type of cloud computing service allows for the development of applications and services.
 - Subscribers need not buy and manage the software and infrastructure underneath it but have authority over deployed applications.
- **Software-as-a-Service (SaaS)**
 - This cloud computing service offers application software to subscribers on-demand over the Internet.

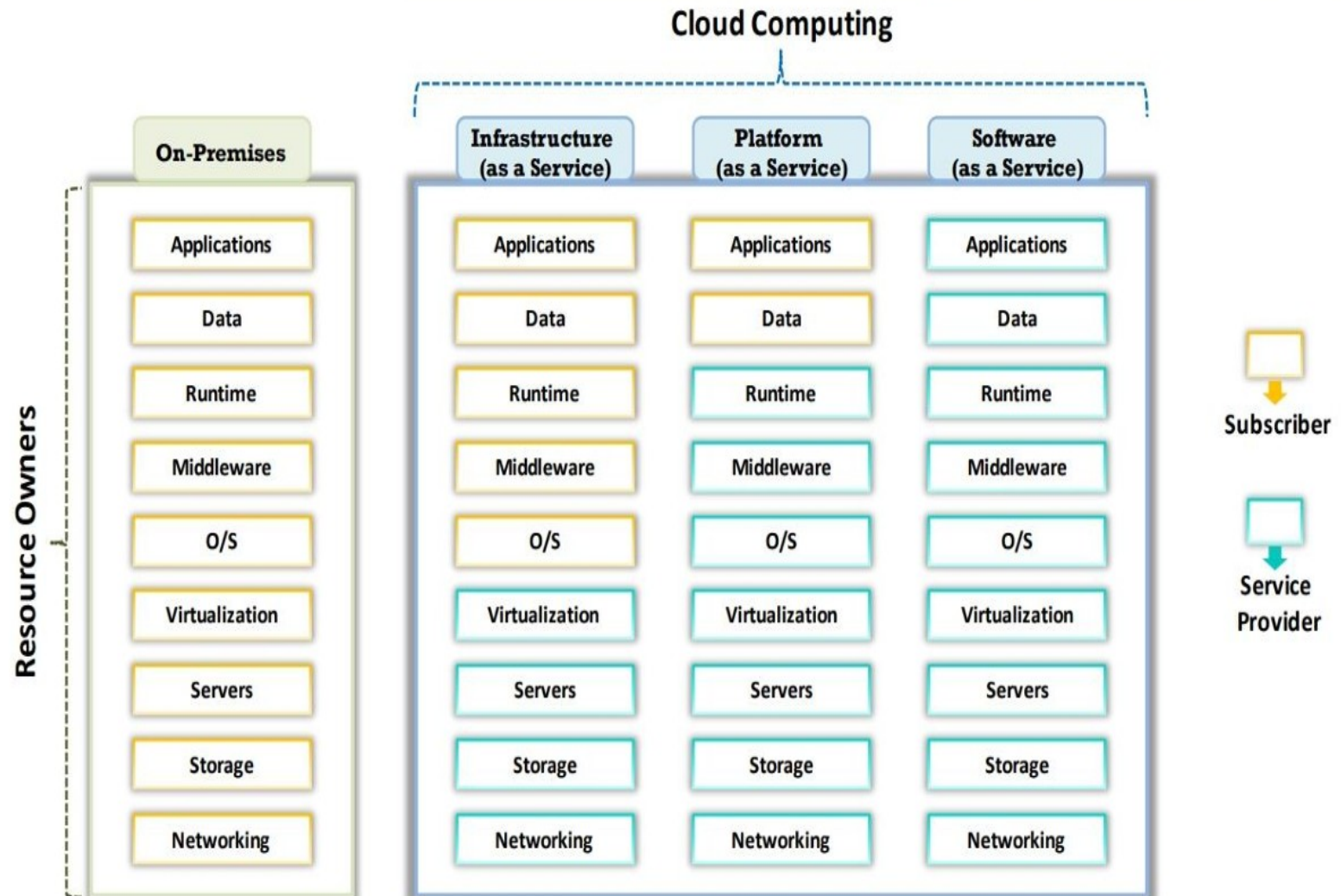
Types of Cloud Computing Services

- Identity-as-a-Service (IDaaS)
 - This cloud computing service offers authentication services to the subscribed enterprises and is managed by a third-party vendor to provide identity and access management services such as Single-Sign-On (SSO) and Multi-Factor- Authentication (MFA).
- Security-as-a-Service (SECaaS)
 - It is developed based on SaaS.
 - It drastically reduces the cost compared to that spent when organizations establish their own security capabilities.
 - It provides services such as penetration testing, authentication, intrusion detection, anti-malware... etc.

Types of Cloud Computing Services

- **Container-as-a-Service (CaaS)**
 - Using these services, subscribers can develop rich scalable containerized applications.
 - CaaS inherits features of both IaaS and PaaS .
- **Function-as-a-Service (FaaS)**
 - This service provides a platform for developing, running, and managing application functionalities without the complexity of building and maintaining necessary infrastructure (serverless architecture).
 - It provides data processing services, such as Internet of Things (IoT) services.
- **Anything-as-a-Service (XaaS)**
 - other types of services such as food, transportation, and medical consultations.
 - XaaS offers secure services such as customer relationship management.

Responsibilities in Cloud



Cloud Deployment Models

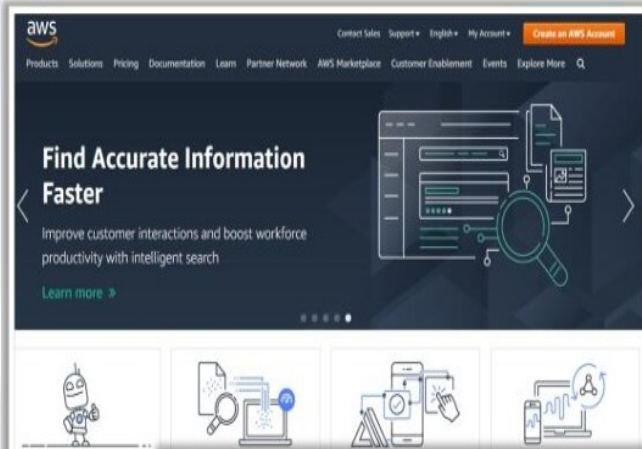
Cloud deployment model selection is based on enterprise requirements.

- **Public Cloud**
 - open for public use.
- **Private Cloud**
 - For single organization only.
- **Community Cloud**
 - several organizations from a specific community .
- **Hybrid Cloud**
 - Combination of two or more clouds (private, community, or public) .

Five major actors in Cloud Computing

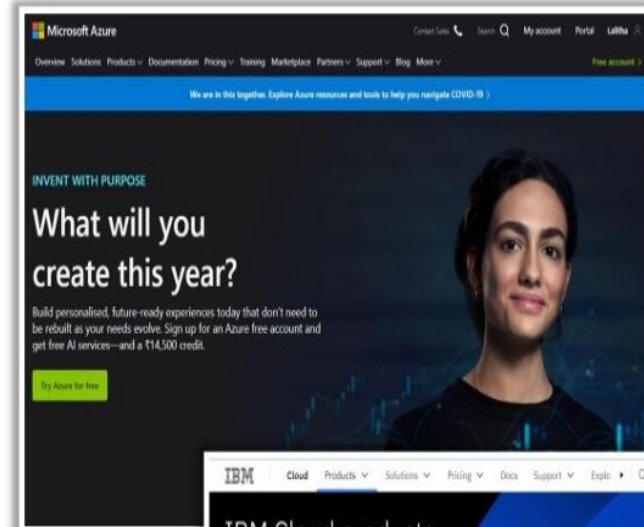
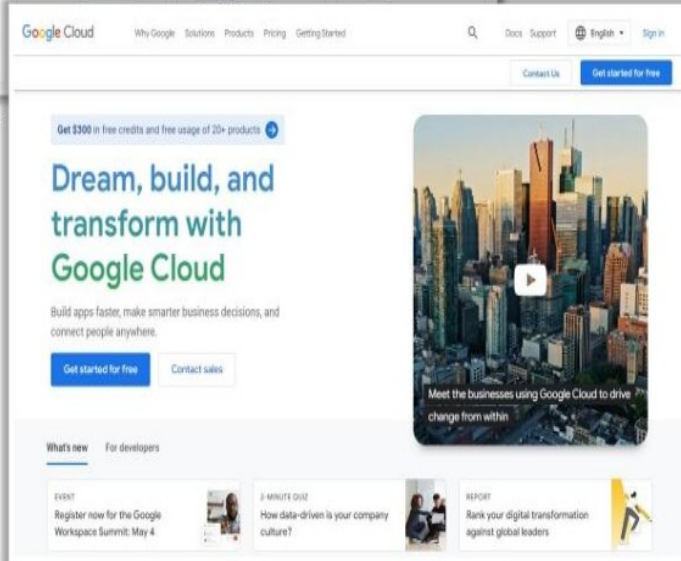
- **Cloud Consumer**
 - A person or organization that uses cloud computing services.
 - sets up service contracts with the CSP(Cloud Service Provider).
- **Cloud Provider**
 - A cloud provider is a person or organization who acquires and manages the computing infrastructure intended for providing services.
- **Cloud Carrier**
 - acts as an intermediary that provides connectivity and transport services between CSPs and cloud consumers via a network.
- **Cloud Auditor**
 - A cloud auditor is an independent entity, person, or automated tool that evaluates and verifies cloud computing services, information, performance, and security controls.
- **Cloud Broker**
 - The cloud broker is an entity that manages cloud services regarding use, performance, and delivery and maintains the relationship between CSPs and cloud consumers.

Cloud Service Providers

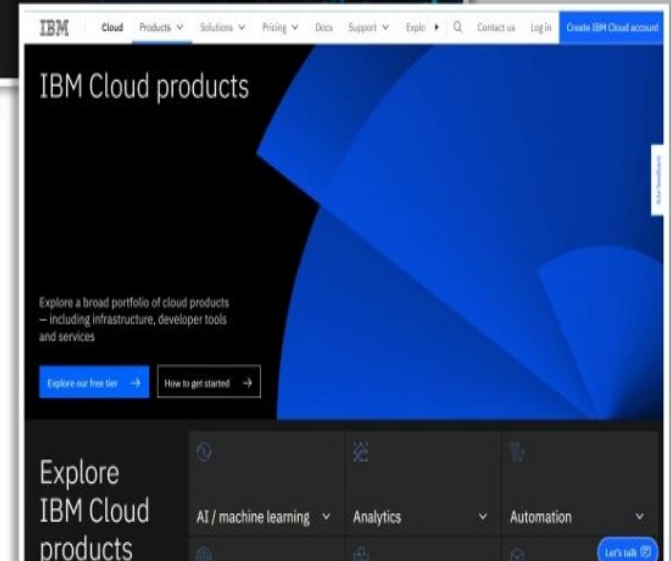


Amazon Lightsail Containers
A new easy-to-use containers service for a low, predictable price

<https://aws.amazon.com>



<https://azure.microsoft.com>



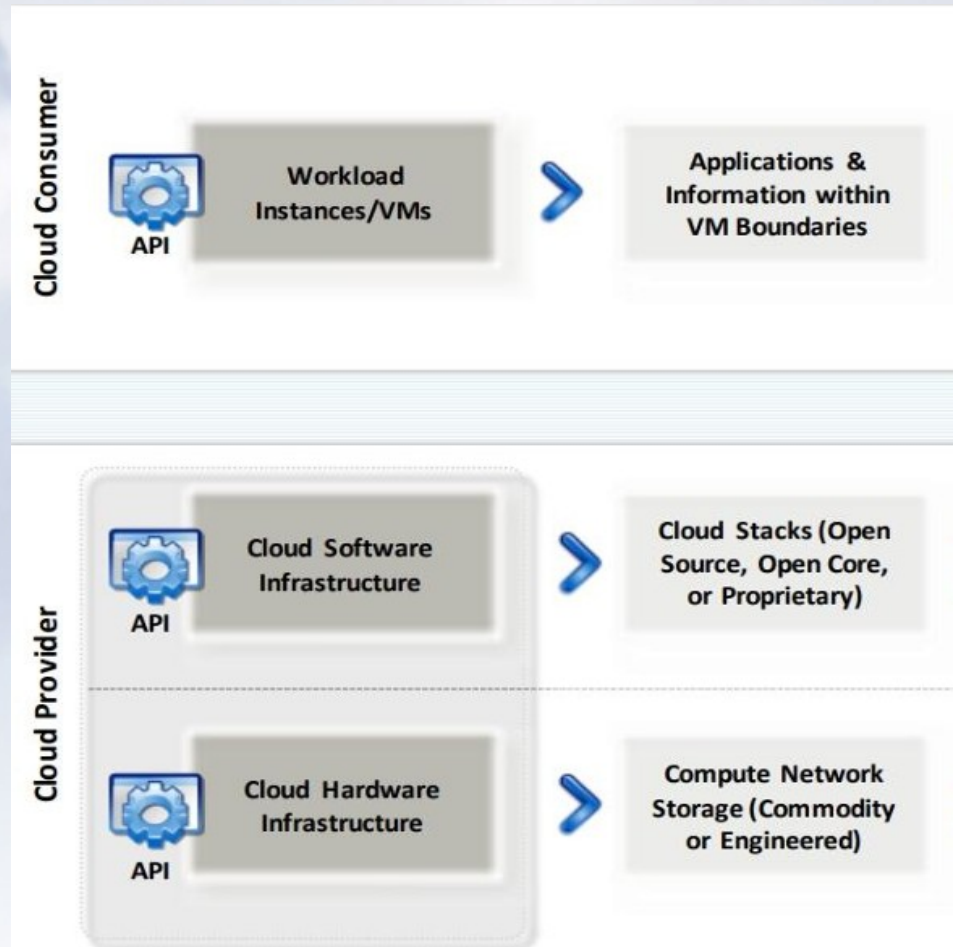
Cloud Computing Threats

- Data breach/loss
 - Data is erased or modified.
 - Encryption keys are lost.
- Unsynchronized System Clocks
- Loss of Operational and Security Logs
- Illegal Access
 - Weak authentication and authorization control to the Cloud.
- Natural Disaster
- Hardware Failure
- Lock-in
 - migrating from one cloud provider to another .

Cloud Security Considerations.

Cloud security is a shared responsibility between the cloud service provider and the customer.

- **Security OF the Cloud**
- **Security IN the Cloud**



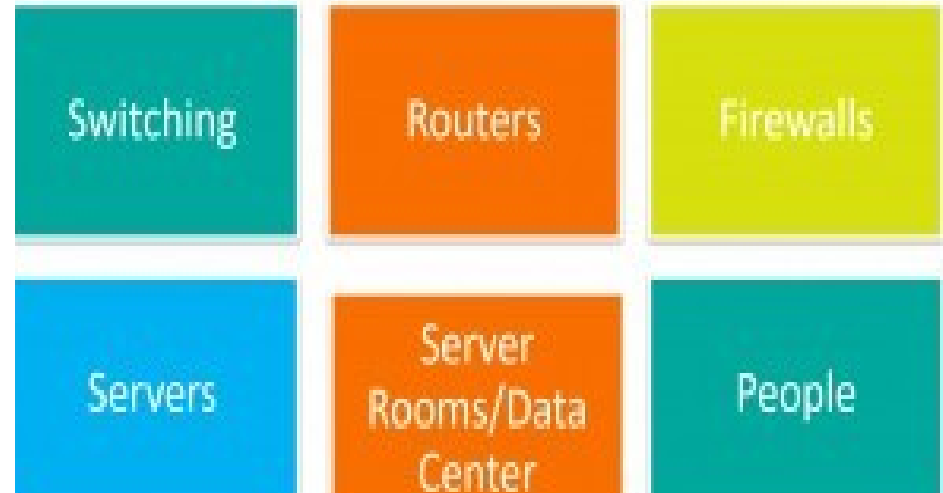
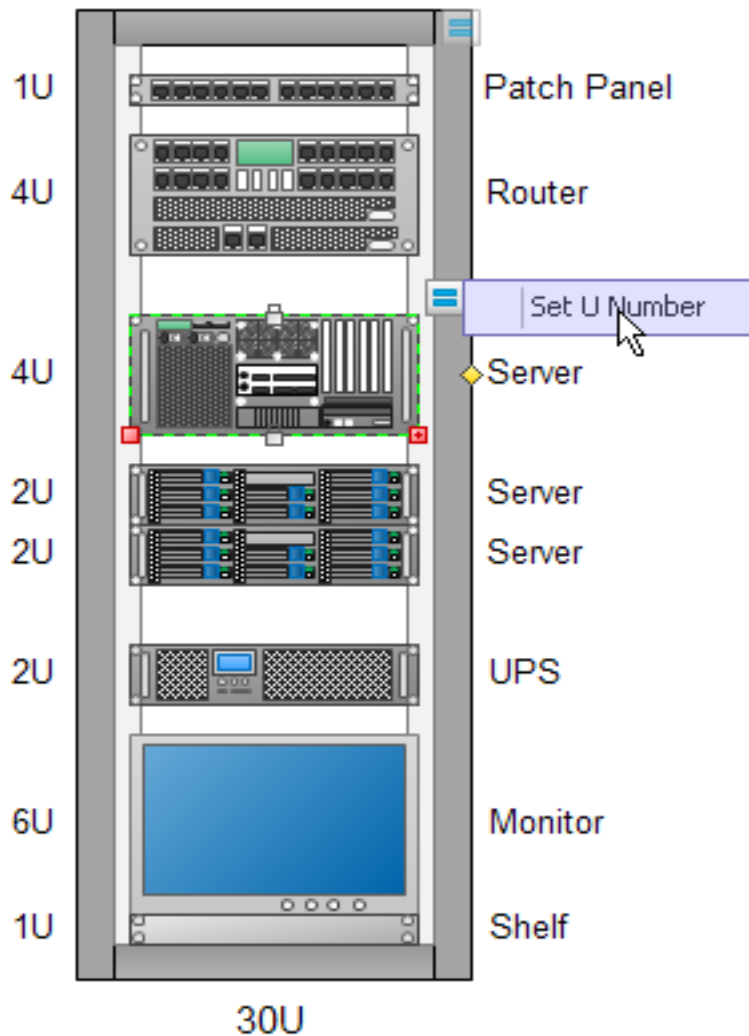
Cloud Security Considerations.

- Cloud services should implement a **disaster recovery** plan.
- Continuous monitoring on the **Quality of Service (QoS)** is required to maintain the service level agreements between consumers and the service provider.
- Data stored in the cloud services should be implemented securely to ensure **data integrity**.
- Cloud computing services should **be fast, reliable**, and need to provide **quick response** times to the new requests.
- **Cryptographic** algorithms must be implemented for optimum data security.
- **Load balancing** should be incorporated into the cloud services.

Cloud Jobs

- **Cloud Engineer**
 - Builds and manages cloud infrastructure, handles migrations, and optimizes performance.
- **Cloud Architect**
 - Designs high-level system blueprints and selects frameworks and cloud services.
- **DevOps / Cloud Operations Engineer**
 - Automates deployments, configures monitoring, and maintains reliable cloud environments.
- **Cloud Security Engineer:**
 - Configures access controls, manages encryption, and protects data against cyber threats

Datacenter Components



Questions



THANKS